SKGArchae: Semantically Inclined Knowledge Graph Synthesis Framework for Archaeology

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Abstract. This paper presents a strategic framework for knowledge graph generation with archaeology as the domain of choice, focusing mainly on its increasing socio-economic importance and limited integration into Web 3.0 systems. This paper uses a framework, which mainly uses GPT-4o for generating auxiliary knowledge, strategically extracting terms and categories from various archaeological datasets, including documents, web sources, and various archives. Latent Dirichlet Allocation (LDA) and a Bi-LSTM are used for generating seed knowledge and classifying metadata respectively (while retaining top 50% instances). The approach is knowledge-repository driven and verified knowledge is emphasized for knowledge graph generation. The quality is maintained through situational quantitative thresholds and statistical measures that improve the pipeline. Further, the hunting search algorithm, with its high heuristic nature, helps in recursive filtering from the initial to the optimal solution of the knowledge graph. A feedback-based and tech sample–based entity mechanism is also used to expand and populate the graph. Archaeology archives from verified documents are included to create domain-specific enrichment. Overall, the arrangement provides a semantics-oriented and recursive approach for generating knowledge graphs in critical areas like archaeology.

Keywords: Archaeological, Knowledge Graph, Semantic AI

1. Introduction

The way of creating and accessing knowledge has certainly changed in the recent times, this is mainly because of the growth of various digital resources in every domain. And in fields like archaeology, a lot of data is in the form of research papers, e-books, archival records, and curated resources on the web. While some sources offer immense potential for knowledge discovery, they still include challenges related to scale, heterogeneity, and contextual interpretation in their fields. Also, many tradi-

tional methods of organizing such data often led to failure as they are unable to capture the deeper semantic relationships, thus, making it difficult to maintain structure and intelligence in these systems.

With Web 3.0 rising, many metadata-driven services and semantic technologies play a central role, it has been seen that there is an urgent need for certain frameworks that can systematically transform raw digital content into semantically enriched knowledge representations. For Web 3.0 to fully evolve into a semantic web, it is essential that metadata be translated into meaningful graph-based structures that can support both human interpretation and machine reasoning. In this process, qualitative reasoning plays an important role by ensuring that instances are populated not only on statistical grounds but also through contextual strength, which reduces unnecessary learning and redundancy while retaining semantically valuable indicators.

One powerful paradigm that has emerged is the use of Knowledge graphs, these represents structured relationships among entities, thus enabling advanced querying, reasoning, and knowledge integration processes. However, in the case of archaeology, existing implementations remain limited. The domain continues to lack the specialized models that can utilize semantic AI, deep learning, and optimization techniques to synthesize coherent knowledge structures from any diverse data sources. This paper mainly addresses this gap by presenting a semantically inclined knowledge graph synthesis framework for the archaeological domain. The proposed approach leverages topic modeling, sequence-based classification, and metaheuristic refinement to construct a scalable and contextually accurate knowledge graph. By formalizing archaeological knowledge into such structures, the framework strengthens the foundation for intelligent information systems while also contributing to the broader development of Web 3.0-compliant knowledge infrastructures all throughout.

Archaeology poses distinctive challenges for knowledge graph construction. Unlike domains such as biomedicine or finance, archaeological data is fragmented across excavation reports, inscriptions, and archival records, with incomplete or inconsistent metadata. Interpretation often varies with cultural and temporal context, and terminological overlaps complicate semantic alignment. These characteristics make archaeology uniquely difficult for automated knowledge synthesis, thereby requiring a domain-tailored framework.

Motivation: Web 3.0 is a sign of transition to dynamic representations of knowledge, rather than the previous static system based on the use of keys. However, the current digitization movement has failed to address the old problem of systematizing the heterogeneous data in the field into systematic structures. With the implementation of the latest modeling methods, it is possible to capture the archaeological information as interdependent networks that maintain the authenticity but, at the same time, make the information searchable and more analytical. The approach to methodology, in its turn, tries to address the gaps in modern information systems that do not only fail to be comprehensive, but are simply not present in the modern epoch of the data world. The tools that are available usually force researchers to collect fragmented pieces instead of dealing with interconnected networks.

Contributions: The offered framework includes a number of new contributions. First, it provides a dual perspective mechanism of archeological knowledge synthesis,

integrating topic modelling (LDA) and natural-language summarization to produce a meaningful seed corpus, which is further streamlined with classification using a bidirectional long short-term memory (Bi-LSTM) model. This is aimed at selecting only the top 50 percent of the cases making it relevant to domains and at the same time achieving scalability. Second, the framework methodically introduces archeological archives and plenty of curated knowledge stacks and approved documentation into a staged storehouse that mainly promotes the incremental development by semantics-oriented logic. Normalized Pointwise Mutual Information (NPMI) is also used which serves to ensure proper population and control of the knowledge graph and reduce noise. The last stage, optimization phase, embraces the meta-heuristic refinement approach that is most appropriate to transform the initial findings into the best-in-class solution to the highly scalable representation of archaeological knowledge.

Organization: This paper follows a systematic structure in a way that Section 2 is an essential list of related works which helped to the research work using reference. Section 3 includes details about the proposed system design. In Section 4, the results and performance evaluation are shown that are generated by the suggested mode. Section 5 ends the paper by going over the practical applicability of the model across a range of industries and their real-world implications.

2. Related Works

Qun et al. [1] has proposed a Tibetan archaeological knowledge search system by making a knowledge graph of cultural relics. Their framework combined with a TB-CRM model has also enabled them to retrieve information with a very high-precision rate. The study mainly emphasized the role of Neo4j and Python-based pipelines in supporting both heritage preservation and scholarly research. Felicetti et al. [2] introduced the concept of Heritage Digital Twins (HDTs), which is basically integrating knowledge graphs and artificial intelligence for cultural heritage preservation. Their work mainly demonstrated how ontologies and AI can synergize to build intelligent systems for interpreting cultural data, thus laying the foundation for self-expanding semantic infrastructures. Faralli et al. [3] has developed a large, interlinked knowledge graph for the Italian cultural heritage which connects entities defined in the ArCo ontology alongside widely used bases such as DBpedia and the Getty Vocabulary Program. Their methodology is mostly based on semi-automatic resource creation for the scalability of linked data for cultural knowledge integration. Handzic and Dizdar [4] also explored the role of knowledge management in archaeology by mainly emphasizing on its potential to facilitate discovery of knowledge. Their work has also highlighted the intersection of big data analytics with knowledge management strategies for supporting archaeological research. Gatiglia [5] has mainly provided information about various applications of artificial intelligence in the domain of archaeology by stressing heavily on the challenges of managing large-scale data. The study has also highlighted the risks of oversimplification mainly when applying AI to archaeological contexts, also it emphasized the need for more richer metadata and interpretative frameworks. Xiong et al. [6] also presented a multi-modal knowledge graph for

processing Oracle Bone Inscriptions. It was achieved by integrating multi-source data, their model thus improved detection accuracy by 3.7% and recognition accuracy by 14.8%. It mainly showcased how multimodal integration enhances both recognition and interpretation of ancient texts. Prignano et al. [7] examined archaeological similarity networks, their study has mainly highlighted the methodological challenges of mapping archaeological data into networks while also showing on how network approaches can truly reveal the underlying structural similarities between various contexts. Hacigüzeller et al. [8] assessed the role of structured digital data in archaeology, analyzing its implications for information, knowledge, and wisdom management. Their work has mainly examined the risks of overemphasis on structured forms, thus, stating about their marginalized contextual richness and interpretative depth. Travé Allepuz et al. [9] proposed ontology-mediated historical data modeling for the purpose of integrating archaeological and historical sources. Their approach has mainly emphasized on both the theoretical and practical aspects of ontology, thus highlighting its potential for more coherent digital heritage infrastructures. Lastly, Felicetti and Niccolucci [10] has expanded on the applications of artificial intelligence and ontologies in managing Heritage Digital Twins data. Their study has strongly pointed out the predictive capabilities of structured ontologies, and AI-enhanced functions which are helpful for minimizing errors such as hallucinations in cultural heritage data management.

3. Methodology

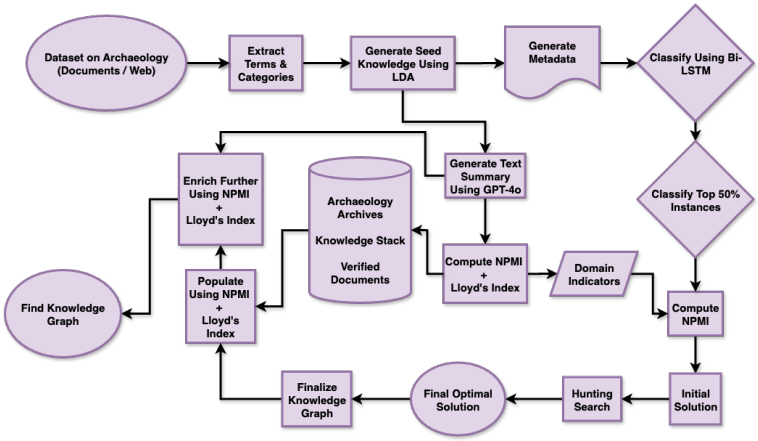


Fig. 1. Model Architecture for Proposed SKGArchae Framework

Figure 1. depicts a strategic knowledge graph synthesis framework for archaeology as the chosen domain. Archaeology was selected because of its growing popularity, yet its integration within Web 3.0 remains minimal. Although numerous documents and research work exist, the incorporation of archaeological knowledge into Web 3.0 strategic metadata and knowledge graphs is still underdeveloped. Hence, formalizing

and building knowledge graph services for archaeology is of utmost importance. In this framework, the first step involves extracting key terms and categories from archaeological datasets, which include embedded documents, web sources, and research papers. These terms and categories are identified based on their frequency, keyword occurrence, and contextual meaning, using processes like tokenization, lemmatization, and stop-word removal.

After extracting terms and categories from archaeological datasets, the next step is to generate seed knowledge using Latent Dirichlet Allocation (LDA). LDA is employed because it produces highly relevant and specific knowledge representations that are useful for metadata generation. The metadata is then organized using DSpace, which aggregates heterogeneous Web 3.0 metadata at scale. Because of the large and complex nature of this metadata, a Bidirectional Long Short-Term Memory (Bi-LSTM) classifier is applied. Bi-LSTM is preferred since it can capture sequential dependencies, does not require manual feature engineering, and provides reliable classification performance. From this classification, only the top 50 percent of instances are retained to preserve domain significance while ensuring scalability.

LDA was selected over more recent transformer-based topic models due to its interpretability and suitability for sparse, domain-specific corpora. Bi-LSTM was adopted for metadata classification because of its ability to capture long-range dependencies in sequential archaeological terms without manual feature engineering. The Hunting Search metaheuristic was preferred to alternatives such as Genetic Algorithms and Particle Swarm Optimization, as it balances exploration and exploitation effectively and avoids premature convergence, making it particularly robust for complex archaeological datasets.

To supplement this stage, auxiliary knowledge is incorporated from verified archaeological resources, including e-books, glossaries, indexes, and research artifacts, which are collectively stored in an archaeology knowledge stack. This repository is further processed using GPT-4o to generate concise text summaries. The resulting terms, after tokenization, lemmatization, and stop-word removal, are evaluated using Normalized Pointwise Mutual Information (NPMI) and Lloyd's Index. Stringent thresholds are applied, with NPMI set at 0.75 and Lloyd's Index at 0.10, to identify domain indicators that capture the underlying essence of archaeological knowledge. These indicators serve as the foundation of an initial solution, which is refined further by computing NPMI at a higher threshold of 0.85. The refined set is then subjected to a Hunting Search metaheuristic, which iteratively explores the solution space. This process ultimately produces the final optimal solution, where entities with strong NPMI values are linked together as neighbors, resulting in a structured and coherent knowledge graph for archaeology.

The finalized knowledge graph is further enriched through the computation of the Normalized Pointwise Mutual Information (NPMI) and Lloyd's Index, applied to both the archaeological archives and the verified documents curated by domain experts. This step produces a best-in-class knowledge graph, where nodes and instances are continuously populated to improve completeness. Additional entities are incorporated using terms extracted from GPT-4o-generated text summaries, which ensures that the graph integrates both expert knowledge and automatically generated metadata. For

this final stage, the thresholds are set to NPMI ≥ 0.70 and Lloyd's Index ≥ 0.15. These values are intentionally less stringent than earlier phases, since the domain-specific knowledge has already been established at higher thresholds. The relaxed criteria allow broader instance inclusion while maintaining semantic coherence. As a result, the knowledge graph achieves a balanced structure that combines rigorous domain indicators with enriched auxiliary information. The outcome is a comprehensive, scalable, and contextually accurate knowledge graph tailored for archaeology.

3.1. Latent Dirichlet Allocation (LDA)

The Latent Dirichlet Allocation (LDA) is a generative model that is a probabilistic model mostly used in topic modeling large volumes of text data. It makes the assumption that documents are composed of several latent subjects, each in the form of a distribution of words. The latent structures inferred by LDA indicate abstract topics that cut across a corpus facilitating the grouping of semantically similar terms. Since it shrinks the dimensionality and generates readable groups of words which depict contextual significance, it is often applied in text analysis, information retrieval, and knowledge discovery.

3.2. Bidirectional Long Short-Term Memory (Bi-LSTM)

A Bidirectional Long Short-Term Memory (Bi LSTM) network is a type of recurrent neural networks that is directly designed to effectively remember the sequential information compared to traditional RNNs. Vanishing and exploding gradient issues plague the existence of traditional RNNs, massively limiting their ability to represent long-range dependencies.

The architectures of Long Short- term Memory address these limitations by adding special memory cells as well as gating mechanisms that is, the input, forget, and output gates that control the flow of information and ensure contextual integrity across longer sequences. The bidirectional variant takes input tokens in both forward and backward directions, so the model is able to utilize both past and future contextual cues simultaneously. The ability is especially relevant in the activities in which the semantic meaning of a token is mostly predetermined by the surrounding factors. Bi-LSTMs enhance and enrich the representation of the sequential dynamics by combining the forward and backward hidden states.

The advantages of Bi-LSTMs in classification are that they automatically discover complex inter-feature relationships that exist in sequential data without depending on features that are hand-crafted. They also compress discriminative patterns to the raw sequences and project them to target categories. Therefore, Bi -LSTMs make a powerful methodology of use in such fields as text classification, voice recognition, sentiment analysis and bioinformatics. Their ability to model dynamically contextual dependencies is translated into significantly good accuracy and strength in a variety of classification problems.

3.3. Hunting Search

The Hunting Search algorithm is one of the populational metaheuristic optimization algorithm mainly based on the foraging behavior caused by cooperation that is shown by natural predators. It mimics the way hunters find prey by simultaneously exploring and exploiting the solution space. The solution spaces of each candidate are viewed as discrete hunters and by repeated cooperative mechanisms the aggregate starts to alter its search path towards minor steps towards the global optimum.

Adaptive capacity is the main strength of Hunting Search algorithm. It biases the searching of new areas of the search space to reduce the effect of premature convergence and at the same time increasing the intensity of search around good subregions. The two-step approach can be used to circumvent local optima and enable the approach to high-quality solutions. Diversity is brought into the population by the stochastic nature of the algorithm and therefore, the algorithm remains robust in the presence of extremely complex and multimodal optimization problems.

Hunting Search methodology is applied in a wide range of fields, such as scheduling, clustering, engineering design and data mining. It is highly flexible in nature and can be used to solve both continuous and discrete optimization problems. The algorithm provides a scalable and efficient tool to apply to computationally challenging optimization problems by constructively encoding biologically inspired hunting strategies into formal mathematical operations.

4. Results and Performance Evaluation

The experimental corpus comprised approximately 4,800 documents, totaling 2.1 million tokens and 12,000 unique terms. Web crawling and entity extraction aggregated both structured metadata and unstructured text. Preprocessing steps included lemmatization, stopword removal, and alignment with a domain-specific dictionary, while Named Entity Recognition extracted entities such as sites, artifacts, and dating methods.

The experiment was conducted on four different restrictive datasets, each of which is carefully chosen for its relevance to archaeological knowledge representation. The first dataset is the Center for Digital Antiquity's 2014 record by McManamon and Smith, which documents archaeology and the United States Federal Government [11]. The second is Woodall's 2011 preliminary report on the archaeological site survey and evaluation of the Vest Mill Road and Lashmit Drive outfalls [12]. The third is a dataset of archaeomagnetic studies on archaeological sites in Tamil Nadu, India (1990) [13]. Lastly, the fourth dataset is Shawn Graham's 2016 Mapping the Structure of the Archaeological Web [14].

These datasets were strategically reformulated for the experimentation using a web-crawling and entity-extraction process. All documents, webpages, and associated terms were aggregated as archaeological datasets within a Web 3.0 environment. The crawling stage allowed us to bring together both structured metadata and unstructured text, which were later aligned through various annotations based on the key terms and domain-specific indicators. Every dataset was therefore not only used in its original

form but also enriched and then normalized, thus, producing a solid experimental base on which the proposed framework was evaluated.

Evaluation protocol: All models were run in the same environment on the same corpus. Each entity and relationship produced by a model was compared with a reference list of archaeological entities and relationships prepared in advance, and checked against the verified expert documents, and the outcome was confirmed by manual review. Precision, recall, accuracy and F-measure were computed from these judgements, with the false discovery rate (FDR) reported as an auxiliary measure. For Fig. 2, precision was measured among the top-k entities returned for a query, for k from 10 to 50.

Table 1. Performance Comparison of the Proposed SKGArchae with other baseline models

Model	Average Precision %	Average Recall %	Average Accuracy %	Average F-Measure %	FDR
T2KB [15]	90.07	91.04	90.56	90.55	0.10
KGCF [16]	91.44	92.85	92.15	92.14	0.09
CogKKG [17]	93.47	94.78	94.13	94.12	0.07
Proposed SKGArchae	96.09	98.41	97.25	97.24	0.04

Performance evaluation of the proposed SKGArchae framework, that is a strategic and semantically inclined knowledge graph for archaeology, was carried out against three baseline models. T2KB denotes the LLM-based, ontology-driven knowledge graph generation baseline of Text2KGBench [15], KGCF denotes the T2KG framework for automatic knowledge graph creation from natural language text [16], and CogKKG denotes the cognitive knowledge graph generation model of Li et al., which combines an attention mechanism with multi-modal factor fusion [17]. The evaluation considered Precision, Recall, Accuracy, and F-Measure as the primary metrics, with False Discovery Rate (FDR) as an auxiliary metric. Precision was mainly chosen because it measures the relevance of the retrieved results, while FDR quantifies the number of false positives yielded by the framework.

From Table 1, the proposed SKGArchae model has given the highest overall values. It achieved an average Precision, Recall, Accuracy, and F-Measure of 96.09%, 98.41%, 97.25%, and 97.24%, with the lowest FDR value of 0.04. On the other hand, T2KB [15] has given 90.07%, 91.04%, 90.56%, and 90.55%, KGCF [16] has given 91.44%, 92.85%, 92.15%, and 92.14%, and CogKKG [17] has given 93.47%, 94.78%, 94.13%, and 94.12% across the same metrics. Their corresponding FDRs are 0.10, 0.09, and 0.07. Although the baselines perform reasonably well, none of them match the balance of precision, recall, and overall accuracy shown by SKGArchae.

All the models were run in the exact same environment and on the same dataset, and the results are computed and depicted in Table 1.

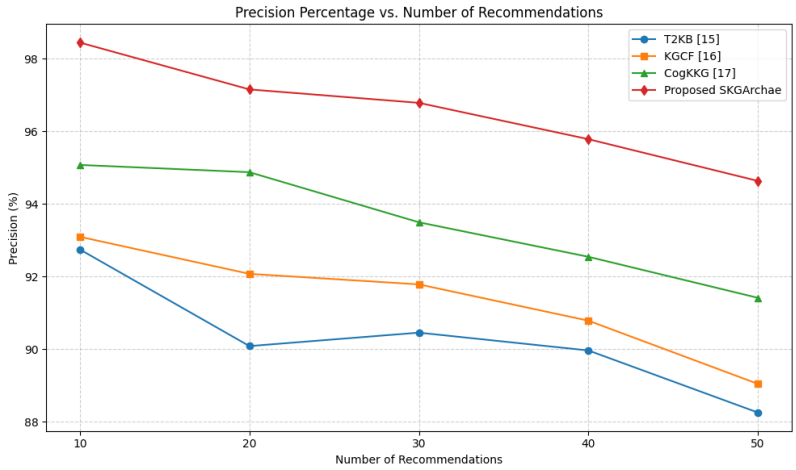


Fig. 2. Precision Percentage vs. Number of Recommendations

Figure 2. depicts the comparison of precision percentage with respect to the number of recommendations. For all the models the precision tends to decrease as the number of recommendations increases, which is expected since larger recommendation sets generally introduce more noise. However, the proposed SKGArchae framework consistently maintains the highest precision across all values of recommendations, starting above 98% and remaining above 96% even at 50 recommendations. The CogKKG [17] model performs better than T2KB [15] and KGCF [16] but still drops more steeply and stays below SKGArchae. The other two baselines, T2KB [15] and KGCF [16], show relatively lower precision and a sharper decline as recommendations increase. This further confirms that SKGArchae is more stable and reliable in retaining precision even as the recommendation size grows.

The reason why SKGArchae is performing better is because it is more domain-centric and designed to incrementally grow knowledge. It extracts important terms and categories from the datasets in different perspectives. The first perspective is generating a seed knowledge using Latent Dirichlet Allocation (LDA), which gives very relevant topics. The second perspective is automatically capturing structural metadata from Web 3.0, which is then classified by a Bi-LSTM classifier. At the same time, concise text summaries are generated using GPT-4o, which also enriches the terms. Most importantly, a knowledge repository comprising archaeological archives, verified documents, and contributions from domain experts is used as a knowledge accelerator. This helps intensify the knowledge at every stage of the pipeline.

A series of semantics-oriented reasoning steps are also applied. Measures such as Normalized Pointwise Mutual Information (NPMI) and Lloyd's Index are used to filter out the best class entities. This ensures that only semantically strong entities remain in the graph. The Hunting Search metaheuristic is then used as an optimization

step, which explores the solution space and refines it until it converges to the best possible solution. Finally, the graph is enriched again with GPT-4o summaries, so that the final output is both formalized and aligned with Web 3.0 standards.

The baseline models have certain limitations which explain their lower performance. T2KB [15] depends heavily on ontology and large language model outputs. It maps sub-concepts and relations, but because it does not include structural metadata, many semantic terms are missed or neglected, and hence it cannot perform as per the SKGArchae model. The KGCF [16] model is based on a hybrid composition that combines rules and similarity-based approaches. It does well in mapping predicates, but incremental addition of knowledge from LLMs and from the verified repository is absent. Meta-transformation and quantitative reasoning are also missing, which means that although it performs moderately, it lags the proposed framework. The CogKKG [17] model uses an attention mechanism with multi-factor fusion and an autoencoder for multimodal features. These are strong techniques, and they do help in fusion, but the model has a dataset-level limitation. It uses only limited auxiliary knowledge and does not incrementally enrich the knowledge graph. Semantics-oriented quantitative reasoning and optimization through Hunting Search are absent, and because of this SKGArchae still performs much better.

Overall, since SKGArchae combines LDA, metadata classification with Bi-LSTM, semantic filtering with NPMI, GPT-4o summaries, and refinement through Hunting Search, it provides a complete and balanced solution. Together with the verified archaeology archives, the framework can be considered a best-in-class model for semantic knowledge graph construction in archaeology, and a prospective approach for building future meta-historical infrastructures.

5. Conclusion

This paper has proposed a strategic and novel framework for knowledge graph synthesis in the domain of archaeology by leveraging many Web-based resources, including documents and many archival records, to generate semantically enriched knowledge structures. A staged, incremental addition of knowledge is used by integrating topic modeling (LDA), large language models like GPT-4o, and deep learning frameworks into the pipeline. A Bi-LSTM network is used to handle the metadata generation and classification to make sure that top 50% of classified instances are retained, thus, ensuring both domain relevance and scalability. GPT-4o does play a dual role in the framework by generating concise summaries that enrich entity representation and providing complete domain indicators for the knowledge graph. Verified archaeological archives and curated documents are systematically incorporated, hence making the framework both domain-specific and reliable. Statistical measures such as Normalized Pointwise Mutual Information (NPMI) and Lloyd's Index are applied at multiple stages to ensure accurate population and semantic coherence. Finally, the Hunting Search metaheuristic, guided by a customized objective function and empirical thresholding, refines the solution space to achieve an optimal and robust knowledge graph. Overall, the proposed SKGArchae framework demonstrates its capacity to

outperform baseline models in both accuracy and scalability, offering the best solution for synthesizing archaeological knowledge graphs.

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